Problem Statement No.22: Smart IoT Device Troubleshooting Chatbot

The Challenge

IoT devices are widely used in smart homes, industries, and connected environments. However, users frequently encounter problems such as connectivity failures, configuration errors, firmware issues, and device compatibility challenges.

Troubleshooting these problems often requires technical knowledge and the ability to interpret complex documentation. Many users depend on manuals, forums, or customer support, which can delay issue resolution.

The challenge is to develop an intelligent assistant that can **quickly diagnose IoT device problems and guide users through troubleshooting steps**.

The Objective

Build a **conversational AI chatbot** that helps users identify and resolve IoT device issues through natural language interaction. The solution should:

1. **Understand Device Issues :-** Interpret user descriptions of IoT problems such as network connectivity failures, configuration errors, or device malfunctions.
2. **Retrieve Relevant Knowledge :-** Access IoT documentation and troubleshooting guides using a Retrieval-Augmented Generation (RAG) approach.
3. **Provide Step-by-Step Solutions :-** Generate clear instructions for resolving connectivity, configuration, or firmware issues.
4. **Conversational Support :-** Offer an interactive chatbot interface where users can ask questions and receive guidance in real time.

Tech Stack

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models